

time basis, then it's appropriate to have a function that standardizes and normalizes the values automatically. The alternative to this would require users to download sample datasets, perform the necessary preprocessing changes and save it to the model mainframe. This is clearly not an efficient way when working with on time analysis projects.

Another thing to consider is that different stakeholders in a data project can have access to different forms of the same dataset. Before using these datasets for analyses, remove any anomalies. One good example of this can be how users format dates, price values (with symbols) or names (First names and last names). Consistent data formatting takes away these errors so that the entire data set uses the same input formatting protocols.

UNDERSTANDING THE DATA VARIABLES

Teach the system to distinguish different columns and to find ways how work can be reduced in formatting the input values. Going back to the example for first names and last names, if the columns for both are split into separate sections in one table but are combined in another, the system should be able to match these two to avoid redundancies.

Learning SQL and other relational database management systems, one comes across concepts for primary keys and secondary keys which play a great role in connecting numerous tables. A primary key is nothing but a column that contains a unique value for each record in the dataset. Make sure that your datasets have at least one primary key column to help in pivoting and combining the inputs. For continuous variables, make sure to use histograms to review the distribution of your data and reduce the skewness. This is done to identify any outliers in the mix that may not contribute much to the model or the analysis results as a whole. Duplicates and redundant values too can diminish the quality of the research, so have them removed well in advance.

What about the pesky problems of missing values? Well there are a host of options that users can avail from. If the number of missing values is insignificant to the total number of distinct records, you could choose to simply remove all the records with incomplete data. But if the missing values are prevalent throughout, using interpolations with average or median values is the best option to go with. Take note, that using mean column values to populate missing cells can change the nature of the distribution graphs and charts.